PHYSICS MODULO GAUGE

The Methodology of the Incompleteness Framework

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ABSTRACT

This paper examines the methodology of a finite reversible observation framework. [Main] establishes an exact finite observable-law equivalence with fixed-basis unitary/Born representations and separately scopes the coherent operational lift. The Standard Model and horizon constructions use additional physical premises on selected branches. Five themes organize the discussion: the distinction between the foundational observation axioms; conditional structural results and specified observational invariances; the hypotheses used in analyses of Bell, Kochen–Specker, Pusey–Barrett–Rudolph and Frauchiger–Renner no-go results; a programme of physics modulo a specified observational equivalence; and conditional operational arguments separating model-derived quantities from empirical selections. Theorem 23 supplies selected-branch outputs, while Theorem 24 distinguishes coherent-channel, population-only and all-partitions comparisons. The restricted coherent-channel generators are non-exhaustive, and the general physical branch selector and observational classification remain open. The methodological proposal does not supply a unique inverse reconstruction or a complete ontology quotient.

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1. INTRODUCTION

1.1 What this paper is, and what it is not

This is a paper about method. The Incompleteness framework is a body of physics — it makes quantitative, parameter-free retrodictions across the Standard Model (including a value matching the JUNO neutrino measurement at 0.07σ) and recovers the Bekenstein–Hawking entropy coefficient as a consistency relation of its horizon calibration — and that physics is developed in a sequence of technical papers — [Main], [SM], [Substratum], [GR] — to which this one is a companion. The present paper does not re-derive any of it. It asks a different question: *what is the framework's methodology*, and is that methodology defensible and replicable independently of whether the framework's particular physical claims survive?

The question is worth separating because the framework's methodological posture is unusual in ways that a reader assessing the physics might not pause over, but that a reader in foundations or philosophy of physics will recognize

as substantive. The framework claims a single foundational commitment and derives the rest. It claims that what other programs treat as free assumptions are, in its setting, forced. It claims a determinate relationship to the major no-go theorems of quantum foundations. And it claims that a long-standing question in the metaphysics of physics — the realism debate — is, as usually posed, the wrong question. Each of these is a methodological claim, evaluable on its own terms, and each is the subject of one part of this paper.

What this paper is not: it is not a defense of the framework's physics, which must stand on the technical papers and their empirical record; it is not a survey of the foundations literature, though it engages that literature where engagement is due; and it is not a finished statement. Part I in particular presents a recent distillation of the framework's foundational commitment, offered for scrutiny rather than asserted as settled. We mark developmental status explicitly wherever it applies.

1.2 The five commitments

The paper is organized around five methodological commitments, developed in Parts I through V.

Part I — the foundational commitment. The framework is often described as resting on a single axiom: *observation occurs*. Part I examines that phrase directly and finds it resolves into two axioms, not one — an indubitable claim that tokened differentiation occurs, and a substantive further claim that the differentiation recurs. Six of the framework's apparent further assumptions are recovered as lemmas of the first axiom; the second axiom is shown, by a no-go result, not to be derivable from the first. The methodological point is the discipline this exhibits: a foundation stated at the exact granularity of what it commits to, with the single non-evidential posit named and isolated. One element of the distillation is a commitment made explicit where it arises (§5): the observer's embeddedness — the proper-part decomposition $V \subsetneq S$ — is taken to belong to the primitive itself, because the framework reads "tokened differentiation" perspectivally (a registering from a locus, against a remainder). The thin alternative, on which embeddedness must instead be added as a third axiom, is noted there; the choice is presentational rather than empirical, and the two-axiom presentation reflects the adopted perspectival reading.

Part II — constructive structural realism. Ontic structural realism holds that physical structure, not individual objects, is what is fundamental and what is known. In OI, the selected cubic branch yields specified lattice/gauge outputs under its explicitly stated hypotheses, while the four proposed substratum transformations do not yet classify full observational equivalence. Pair C refutes exhaustiveness for restricted fixed-interface coherent-channel generators, and the all-partitions equivalence is open. H-Bell conditions Bell-inclusive existence and full-completion uniqueness remains open. The methodological aspiration of a complete structural quotient is therefore a research objective, not a completed

theorem.

Part III — per-theorem evasion of no-go results. The no-go theorems of quantum foundations — Bell, Kochen–Specker, the reality theorems, Frauchiger–Renner — each constrain hidden-variable accounts of quantum mechanics. The framework is a deterministic-substratum account and must answer to them. Part III shows the answer is not four separate escapes but one structural feature: each theorem assumes, at the level of the hidden substratum, a property that in the framework holds only at the level of the observable emergent description, and the trace-out construction that relates the two levels is what produces the failed assumption in each case.

Part IV — physics modulo gauge. The three preceding commitments combine into a general proposal. The realism debate asks what is fundamental; the framework's posture is that this question is ill-posed, and that the well-posed replacement is: what is the equivalence class of structures that produce the observations, and what are its representatives? Part IV articulates this as a methodology for fundamental physics generally, not as a feature peculiar to the framework's particular substratum.

Part V — the operational methodology. Parts II–IV establish physics modulo gauge as the right form of the answer; Part V develops what the methodology determines operationally. Two further no-go results sharpen the boundary between substrate-derivable content and content selected by empirical input: a gauge-invariance no-go on substrate-only computation, and a category-error finding on substrate-level Lagrangians. These two results unify with Part III's four cases under a single architectural pattern. Systematic audits applying the operational standard to the framework's own content are summarized, and three methodological refinements canonized by the audit experience are stated at the generality the methodology requires.

1.3 Developmental status, stated once

The five parts have distinct scopes. Part I examines the observation axioms and their proposed no-go relation. Part II summarizes conditional selected-branch outputs, exact carrier-specific invariances and the open inverse/classification problems in the technical papers. Part III examines the stated no-go cases under their respective hypotheses. Part IV proposes physics modulo a specified observational equivalence. Part V gives conditional operational arguments and audits of model-dependent versus empirically selected content. These are methodological analyses; they do not replace the physical selection or equivalence proofs they identify as missing.

One status distinction inside the organized physics is worth stating here, because the parts are easy to run together. On the bridge between embedded observation and quantum mechanics, three things hold at three different strengths. Exact finite observable-law representability is **established**: every finite-horizon visible law has a finite reversible realization and a fixed-basis unitary/Born rep-

resentation, and the three descriptions are equivalent. Exact finite operational quantum mechanics is **characterized conditionally**: it is equivalent to the framework's core together with explicit added principles, none of which bare observation incompleteness supplies. Selection of a unique relative quantum evolution from the coherent lifts compatible with one visible family is **open**: ordinary coherence is proved insufficient, no mechanism is endorsed, and nothing in the framework claims the selection is impossible.

PART I — THE FOUNDATIONAL COMMITMENT

Part I develops the framework's foundational commitment: the distillation of "observation occurs" into a two-axiom base, the recovery of six apparent assumptions as lemmas, and the no-go result that isolates the single non-evidential posit. It is the first of the paper's four methodological commitments — the discipline of stating a foundation at the exact granularity of what it commits to.

2. THE EVIDENCE

2.1 What cannot be doubted

The framework invokes the Cartesian move: there is a fact that cannot be doubted, because the act of doubting it instantiates it. It is worth stating that fact in its most compressed form, because the compression is not the form the framework's definitions use, and the gap between them is the subject of this note.

The indubitable fact is **not** "I exist." "I" is a substantial noun; it imports a self, a persisting subject, a bearer of properties. It is not "an observer exists," for the same reason — "observer" is a relational substance-noun, importing both a thing that observes and a persistence through which it does so. It is not "the world exists" — that imports a totality.

What cannot be doubted, stated without importing any of these, is that **there is differentiated content**. There is some *this rather than that*. The registered content, whatever else is true or false of it, is not blank, not homogeneous: it has distinction in it. We will call this *tokened differentiation* — "differentiation" because the content is differentiated, distinction-bearing; "tokened" because what is in question is a concrete occurrence, an instance, not the abstract truth of a proposition. The abstract fact that seven differs from eight does not *occur*; what occurs is the concrete registering of a this-not-that.

This is the evidence. It is genuinely indubitable: to doubt it is to entertain the doubted against the affirmed, which is to register a distinction, which is an instance of the very thing doubted. It is also genuinely *thin* — thinner than "I

exist," thinner than "a world exists." It asserts neither a subject nor a totality. It asserts that differentiation has occurred.

2.2 Why not thinner, why not thicker

One might propose a thinner floor still: "something exists." This is in fact *too* thin to be the evidence. Bare undifferentiated existence is not what is ever presented. What is presented is always some *this as against that* — structured content, not featureless being. Remove the differentiation and nothing remains to be aware of or to doubt. "Something exists" understates the evidence: the evidence includes the differentiation, not merely the existence.

One might propose a thicker floor: "the universe exists," "matter exists," "I exist." Each of these is more than the evidence. Each becomes the object of a coherent doubt the moment one asks how it is known; each, doubted, unravels to the thin fact beneath — that there is differentiated content. The thicker claims are not evidence. They are hypotheses about what lies behind the evidence.

Tokened differentiation is therefore the unique level that is *both* indubitable *and* the actual content of what is given: more than bare existence, less than a world, silent on every question of what underlies it. That it is also exactly thick enough to generate structure — as Section 4 shows — is not built in here; it is a fact established downstream, and it is what makes this level the right floor rather than an arbitrary one.

2.3 What the evidence is silent on

It is as important to state what tokened differentiation does *not* deliver as what it does.

It does not deliver a subject: there is differentiation, but the evidence does not say it is differentiation *for someone*, registered *by* an entity that has it.

It does not deliver a totality: there is differentiation, but the evidence does not say it is part of a larger whole, or that there is anything beyond it.

It does not deliver a ground: there is differentiation, but the evidence is silent on whether it is self-grounded, grounded in something else, or grounded in nothing. It asserts the occurrence and stops.

And — the point that will occupy Sections 3 and 5 — it does not deliver *time*. The fact "there is differentiated content" is, as stated, tenseless. It is satisfied by a single differentiated tableau: a this-here against a that-there, with no before and no after. The evidence gives a frame; it does not, on its own, give a sequence of frames.

2.4 The distinction-primitive: lineage and the divergence from it

Taking distinction as the primitive is not new, and the note should be explicit about its lineage. The major precedent is George Spencer-Brown's *Laws of Form*

[SpencerBrown], which takes distinction as the irreducible starting point — "we cannot make an indication without drawing a distinction," and so "the form of distinction" is taken "for the form" itself. Spencer-Brown treats distinction as a proto-operation prior to all calculation and all representation: one cannot even compute $1 + 1$ without first distinguishing the symbols. The note's Axiom 1 is, in its core, the same recognition — that differentiation is the floor beneath which analysis does not reach.

But there is a deliberate and load-bearing divergence, and stating it precisely is part of what Axiom 1 contributes. Spencer-Brown's distinction is an *act*: the calculus begins with the *drawing* of a distinction, the *making* of a mark; in his treatment, as commentators note, there is no separation between the result of distinction and the *process* of distinguishing. That actival, processual reading is exactly what this note's Section 5.2 isolates as declined-attempt (1) — "distinguishing as act" — and rejects, because an act takes time and so smuggles succession into what is meant to be the tenseless floor. Axiom 1 therefore adopts Spencer-Brown's distinction-as-primitive but in a *tenseless, non-actival* form: it asserts that a distinction is *tokened* — concretely instantiated — not that a distinction is *drawn*. The difference is not cosmetic. It is precisely the difference that keeps Axiom 1 free of the temporal commitment that belongs, instead, to Axiom 2. Where Spencer-Brown's single actival primitive already contains the seed of process, the note's two-axiom structure separates the tenseless content (Axiom 1) from the recurrence (Axiom 2) that an act would have fused into it.

This divergence also marks the limit of the borrowing. *Laws of Form* has been read, in the literature it generated, as a basis for claims about cybernetics, consciousness, and non-dualist metaphysics. The note takes from Spencer-Brown only the formal point — distinction as the irreducible primitive — and, consistent with Section 2.3, remains silent on the metaphysical and phenomenological superstructure that others have built on it. Axiom 1 asserts that a tokened differentiation occurs; it does not assert what the literature around *Laws of Form* has often gone on to assert.

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3. THE EIGHT-RUNG LADDER

3.1 The framework's definitions, unpacked

The framework formalizes "observation occurs" through a definition: an observation is a triple (S, φ, V) — a total system S, a dynamics $\varphi : S \to S$, and an observer $V \subsetneq S$, a proper subsystem with finitely many distinguishable internal states, coupled to its complement through φ ([Main] §1.2). Three lemmas and four conditions (C1–C4) are then attached. Laid against the evidence of Section 2, this formalization is seen to assert considerably more than "there is differentiated content." The surplus can be itemized as a ladder, each rung a distinct claim:

1. **A distinction occurs.** Differentiated content; this-not-that. *(The evidence of Section 2.)*
2. **The distinction is relational.** It holds between relata — a this *and* a that.
3. **The relata compose a whole the distinction does not exhaust.** There is an S, and the observer-locus V is a *proper* part of it: $V \subsetneq S$.
4. **There is more than one moment.** Temporal succession exists.
5. **A rule connects the moments.** There is a dynamics φ.
6. **The rule is bijective.** φ is deterministic (injective) and reversible (surjective).
7. **The states are finite in number.** The visible configuration space is finite.
8. **The hidden-sector conditions hold.** Non-zero coupling, record persistence (the slow bath one realization), sufficient capacity, history readback (C1–C4).

The framework's theorem — that an embedded observer's finite observable law has an exact fixed-basis unitary/Born representation, Markov laws included, and carries positive conditional memory at an accessible order exactly when history readback holds there, the compressed representation holding for processes in the correspondence's class under (T) — takes all eight rungs as its basis. The evidence supplies only rung 1. The question of Sections 4 and 5 is which of rungs 2–8 are *consequences* of rung 1 and which are *additional commitments*.

3.2 The standard of derivation

A rung counts as derived — as a lemma rather than an addition — only if it follows from rung 1 by an argument that does not, at any step, presuppose a higher rung. The constraint is strict because the rungs are not independent: it would be circular to derive rung 4 using rung 5, or rung 6 using rung 7. Each derivation below is checked against this standard, and where a derivation consumes a premise not yet established, that is recorded as a finding rather than glossed.

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4. WHICH RUNGS ARE LEMMAS

4.1 Rung 2 — relationality

A distinction, by what the word denotes, holds between relata. A distinction that distinguishes nothing from nothing is not a distinction; the notion is incoherent without at least two terms. Rung 2 is therefore not an addition to rung 1 but an unpacking of it: "differentiation" already means *differentiation of something from something*. Rung 2 collapses into rung 1 by conceptual analysis.

4.2 The locus, and rung 3 — the proper part

Rung 3 asserts that the observer-locus is a proper part of an unexhausted whole. Between rung 2 and rung 3 there is a step easy to miss: rung 2 gives two relata symmetrically, while rung 3 gives an asymmetric part-whole structure, one relatum *contained in* a whole it does not exhaust. The step requires a premise — call it the *locus* premise — that the differentiation is registered *at* a locus, a where of the registering distinct from the what registered.

The locus premise is not a free addition. It is recovered from the verb in rung 1. To say a distinction *occurs* — as against merely *obtaining*, the way an abstract mathematical inequality obtains — is to say it is instantiated as a concrete token. And a concrete token is located: occurrence is indexical, a token is a *this-here*, instantiation places. The locus premise is therefore analytic in "occurs," just as relationality is analytic in "distinction." Once the locus is granted, rung 3 follows: the locus V, together with the registered content beyond it, composes a whole S of which V is a proper part, because V does not include the content beyond it. Rung 3 is a lemma.

This recovery has a substantive consequence. The proper-part structure $V \subsetneq S$ — the *incompleteness* the framework is named for — is not an independent postulate. It is a consequence of the evidence, provided "occurs" is read as "is tokened." Embeddedness is analytic in occurrence.

This recovery rests on a reading of the primitive, adopted here explicitly. The step from the locus premise to rung 3 depends on reading "tokened" as carrying a *perspectival* locus — a registering *from* a within-place, against a remainder. On a thinner reading, "tokened" means only that the differentiation is concretely instantiated, with no within-place; on that reading the locus premise does not yield a *proper* part, and $V \subsetneq S$ would be a third axiom rather than analytic in the primitive. The framework adopts the perspectival reading — the floor the cogito secures is already a registering — so embeddedness is part of the primitive and the foundation is two axioms, not three. The choice is presentational: it changes no result downstream of the structure (S, φ, V). It is stated here rather than deferred, and rung 3 is presented as analytic in the perspectival primitive, with the thin-reading alternative noted.

4.3 Rung 5 — the existence of a dynamics

Granting (provisionally, pending Section 5) that there is more than one moment, and granting the locus that persists across moments as the same locus, there is necessarily *something* that relates the configuration at one moment to the configuration at the next. That relation is the dynamics φ. Rung 5 — the *existence* of a dynamics — is therefore a lemma, given rung 4 and the locus.

Two cautions. First, the lemma delivers only the existence of a successor relation, not its uniformity: that the same φ governs every locus and every moment is a further claim, recovered downstream from isotropy and homogeneity, not here.

Second, the lemma depends on rung 4, which Section 5 will show is *not* itself a lemma. Rung 5 is thus a *conditional* lemma: a consequence of the evidence together with the one genuine addition.

4.4 Rung 6 — bijectivity

The framework's Lemma 3 derives bijectivity from the preservation of the observer's records. The derivation separates into two halves with different costs.

Determinism (injectivity). A registered distinction, once it has occurred, has occurred. If the dynamics φ merged two distinct total states into one successor — if φ were non-injective — then two configurations carrying distinct registered records would become a single configuration, and a distinction that had occurred would cease to have occurred. Rung 1 asserts the distinction occurs; persistence of the record across moments (rung 4 plus the locus) carries that occurrence forward; non-injective φ would violate it. So φ is injective. Determinism is a lemma, given rung 1, rung 4, and the locus.

The hidden-sector margin, closed. The record argument as just stated covers merges of configurations carrying distinct registered records. A merge of configurations differing *only in the hidden sector* erases no record, and needs a separate argument; there is one, and it is a dichotomy. If the merged configurations' successors differ visibly, the merge leaves a statistical trace: the marginal of a bijection under the uniform prior is doubly stochastic (by counting: each visible value has exactly as many preimages as there are hidden states; the framework's Main paper, §3.2), unitary emergent statistics are unistochastic, and a visibly-asymmetric merge breaks double stochasticity of the marginalized process at some step order — so such merges are excluded where the observed statistics are of that doubly stochastic class under the uniform prior, an evidential input rather than a further axiom. If instead the merge leaves no statistical trace, it is removable: any map on a finite set restricts to a bijection on its eventual image, a recurrent registered history lives in that image, and the non-injective description is gauge-equivalent to a bijective one — the merge is a pre-history transient with no observational content, and the bijective representative is canonical (this is Part IV's surplus-as-inaccessibility, applied to the dynamics itself). Either horn delivers the bijective substratum for the dynamics governing registered history. The honest accounting, which the derivation's purity label should carry: injectivity on record-distinct configurations is a lemma of rung 1; injectivity on the hidden-sector margin is delivered by observed unitarity or by recurrence — a disciplined mixture of the evidential floor with one empirical input, not analysis of the evidential floor alone.

Reversibility (surjectivity). Injectivity yields bijectivity only on a finite set; on an infinite set an injective map need not be surjective. Reversibility therefore is not established here — it waits on rung 7.

4.5 Rung 7 — finiteness

Rung 7 asserts the visible configuration space is finite. The framework's Lemma 1 motivates finiteness from holographic entropy bounds — but those are downstream physical results, and invoking them at the foundational layer would be circular. The question is whether finiteness follows from rung 1 alone.

It does, in the form the framework actually requires. A *token* is a determinate concrete particular. A single tokened differentiation registers a determinate, and therefore finitely structured, content: an *infinite completed totality* of tokened distinctions is not something that can *occur*, because occurrence is the instantiation of a determinate particular and a completed infinity is not a determinate particular. Infinitely many distinctions could be *constructed* — one tokening after another, across moments — but never *given* in a single occurrence. The evidence therefore delivers finiteness *per moment*: each occurrence registers finitely many distinctions.

Finiteness-per-moment is exactly what the observation base delivers, and the scope of what it delivers should be stated on both of its axes. On the time axis: the visible configuration space $\mathcal{C}_V$ whose finiteness Lemma 1 requires is the space of states distinguishable *at the locus*, a per-moment object; the framework does not require, and should not claim, finiteness of the entire history, and an unbounded number of moments is consistent with the evidence. On the space axis: the tokening argument reaches only the *visible* factor — what the observer registers — so what is recovered as a lemma is per-moment finiteness of $\mathcal{C}_V$. The substratum arguments that act on the *total* per-moment space $\mathcal{C}_V \times \mathcal{C}_H$ — the Poincaré-recurrence step and the eventual-image restriction of the determinism recovery — consume finiteness of the hidden factor as well, and that extension is not delivered here: it is the structural assumption A1 of the substratum construction, counted as such in the companion paper's posit ledger ([Main §4.5], of which this rung analysis is the complementary view — the ladder counts what is recovered from the two-axiom base, the ledger counts what the full argument assumes). Rung 7, in the form Lemma 1 needs, is a lemma.

This completes rung 6 with the same scoping. Determinism (4.4) plus finiteness-per-moment yields bijectivity as a lemma on the visible factor: an injective map on a finite per-moment configuration space is surjective on it. Bijectivity of the substratum map on the total space inherits the A1 extension above; reversibility of the registered dynamics is recovered, and rung 6 is a lemma at the visible level, structural-plus-A1 at the substratum level.

4.6 Rung 8 — the hidden-sector conditions

Rungs C1–C4 are conditions on the relationship between the locus V and its complement. C1, non-zero coupling, follows conditionally: a distinction with content *about* the complement requires coupling to the complement, on pain of that content being unregisterable. C2 (record persistence — slow-bath sep-

aration one route) and C3 (sufficient capacity) are not lemmas — they are substantive conditions satisfied by some partitions and not others. The framework does not claim them as lemmas: it presents C1–C4 as the conditions defining *which* embeddings carry the memory-bearing sector — hidden predictive memory unavoidable in every completion, the quantum representation itself universal ([Main] §3.4) and as a structural selection rule on where observers can exist ([Main] §4.6). Rung 8 is correctly treated by the framework as a selection condition, not a derivation, and no overclaim is present.

4.7 Summary of the recoveries

Of rungs 2–8: rung 2 (relationality) and the locus premise collapse into rung 1 by conceptual analysis; rung 3 (proper part), rung 5 (dynamics, conditionally), rung 6 (bijectivity), and rung 7 (finiteness-per-moment) are lemmas; rung 8 (C1–C4) is a selection condition the framework already treats as such. Exactly one rung remains: rung 4.

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5. THE ONE RUNG THAT IS NOT A LEMMA

5.1 Two questions about succession, and which one is the claim

The claim of this section concerns rung 4 — temporal succession. But "deriving succession" can mean two distinct things, and the no-go applies to one and not the other. The distinction is essential and the rest of this note depends on it.

The first question is the **ordering** question: *given* that there are multiple moments — multiple states of some persisting thing — can the relation "before/after" among them be defined without circularly invoking "before"? This is the problem the philosophy of time has worked on for a century. It is contested but not obviously hopeless: a reductive account of the ordering relation may well be possible (Section 5.3 discusses a recent candidate). The no-go of this note is *not* a claim about the ordering question.

The second question is the **domain-existence** question: is there a temporal domain *at all* — is there more than one moment, more than one state of a persisting thing — and can *that* be derived from a premise which does not already contain it? This is the question the no-go addresses, and the answer is negative.

The distinction matters because the two questions are routinely conflated, and an account that answers the first is easily mistaken for one that answers the second. An account of the ordering relation always *begins* from a set of moments, or a set of states of an object, or a configuration space — a multiplicity given at the outset — and derives the order *on* that multiplicity. It does not derive the multiplicity. The multiplicity is its input. And the multiplicity — that differentiation is instantiated across moments and not only within a single

configuration — is exactly what the framework's second axiom asserts (Section 5.4).

Rung 4, as the framework needs it, is the domain-existence claim: that there is a temporal domain for the dynamics φ to act across. The claim of this section is that this cannot be derived from rung 1. Tokened differentiation, as established in Section 2, is tenseless: it is satisfied by a single differentiated tableau — a configuration, one frame — with no before and no after. To reach even *two* moments from a single differentiated frame requires a premise the frame does not contain. The argument takes the form of a small no-go result.

The two-question distinction also fixes the complete inventory of the framework's temporal commitments. The framework's downstream physics involves more temporal structure than the bare domain-existence claim — it involves an *ordering* of moments, a *causal topology*, and a derived *arrow* of time — and a reader is owed an account of how these relate to the two axioms. The inventory has four levels. *Multiplicity* — that there is more than one moment — is the domain-existence claim; it is the framework's second axiom. *Ordering* — that the moments form a sequence — is the ordering question above; the framework leaves it open. *Causal topology* — that the ordering carries influence-structure, that earlier moments can affect later ones — is what the dynamics requires to be stated at all. *The arrow* — that the ordering has an experienced direction — is derived in the framework's substratum treatment from horizon growth. The relations among these four are definite, and reduce the inventory to the two axioms-or-questions already named. The causal topology is not an independent commitment: an influence-structure on an ordered set of moments connected by φ is nothing over and above the ordering together with φ itself, since "earlier influences later" just means "related by forward iteration of φ"; the dynamics φ is independently present, so the causal topology reduces to the ordering. The arrow is derived, not posited, and rests on the causal topology, hence ultimately on the ordering. So the framework's genuinely independent temporal commitments are exactly two — multiplicity and ordering — and they are precisely the two questions this section distinguishes: multiplicity is settled by the second axiom, ordering is left open. Causal topology and the arrow are downstream of these two and of structure (the dynamics, horizon growth) the framework has independently. No third temporal axiom is required, and none is concealed: the two-question framing of this section is the complete account of where the framework's temporal commitments lie.

5.2 Six attempted derivations

The claim is established not by a single argument but by the exhaustion of the available derivations. Six attempts to derive succession from differentiation were constructed; each fails, and — the point of substance — each fails at a *distinct, identifiable step*, always the same kind of step: the illicit move from a state-predicate to a process-predicate, or from logical order to temporal order.

1. *Distinguishing as act.* "To distinguish is to do something, and doing takes time." Fails: it reads "distinguish" as an act, a process, when the evidence gives only differentiated content statically present. The frame need not be *arrived at*.

2. *Registering as record.* "To register is to inscribe a record, which persists to be read; reading is later than inscribing." Fails: it builds persistence — succession itself — into the meaning of "register." If "register" means only "differentiated content is present," no second moment follows.

3. *Actualizing as transition.* "To be actual rather than possible is to have been actualized, a transition from possibility to actuality." Fails: "transition" and "actualized" are temporal. Actuality can simply *be the case* tenselessly; the space of unrealized alternatives is a contrastive description, not a prior state the token passed through.

4. *Occurrence as actualization.* A refinement of (3) taking "occurrence" as the primitive. Fails at the same step: it still requires actuality to be the *outcome* of an actualizing, and "outcome" is temporal.

5. *Occurrence taking itself as object.* "Occurrence occurs — the occurrence of occurrence — and this self-application unrolls into a sequence." Fails differently: self-application generates a *logical* hierarchy of orders, not a *temporal* sequence. Reading the logical "of" as a temporal "then" is the added premise. (This attempt has a second defect: it imports a self-grounding, *causa sui*, thesis not present in the evidence, trading evidential austerity for elegance.)

6. *A single fused term.* The proposal that some sufficiently broad word denotes axiom and succession together as one. Fails: a single term denoting two separable contents does not unify them; it conceals the seam, so that subsequent users inherit the succession commitment without being told they have made it. Concealment is not derivation.

Six attempts, six distinct points of failure, one invariant: succession is reintroduced, never derived. The consistency of the failure across genuinely different framings is the evidence that the obstruction is structural — a fact about what can be inferred from what — and not an artifact of any one bad formulation. A static premise does not yield a process conclusion; "differentiated content is" does not yield "differentiated content comes to be." In each case what cannot be reached is the domain-existence claim of Section 5.1: each attempt, to get a second moment, must assume a second moment.

5.3 The result against the existing literature

The no-go of Section 5.2 is not made in isolation from a substantial existing literature on temporal succession. That literature both corroborates the result and sharpens the distinction of Section 5.1.

Corroboration — Whitrow. G. J. Whitrow, in *The Natural Philosophy of Time* [Whitrow], surveyed the causal theorists' attempts to define temporal succession — the ordering of events by "before" and "after" — and found that they failed because they "tacitly invoked the very temporal order to be defined." Whitrow's verdict on Reichenbach's mark method, among the most ingenious such attempts, was the same. He concluded that "before" and "after" must be accepted as elementary. "Tacitly invoked" is precisely the diagnosis of Section 5.2: the attempts smuggle. That a major figure in the philosophy of time reached the same finding by independent survey is evidence the obstruction is real and not an artifact of this note's particular six framings.

The ordering question may be tractable — and that does not touch the no-go. It would overstate the result to claim the *ordering* relation cannot be reductively defined. A recent account (Wassermann [Wassermann]) proposes deriving "before" from a timeless "existence asymmetry" on the set of an object's realized states — a genuine candidate for answering the ordering question non-circularly. But the construction begins from M_X, the set of an object's actually-realized states. A set of multiple states of one persisting object is already a temporal domain: one object, many states, is exactly domain multiplicity. The account derives the *ordering* of those states; it assumes the *multiplicity.* This is not a criticism of the account — it answers the ordering question, which is the question it set out to answer. It is an illustration of the Section 5.1 distinction: even the most sophisticated reductive account of "before" takes the multiplicity as its starting set and cannot, by its own construction, derive it.

The same pattern in physics. The timeless approaches to quantum gravity exhibit the identical structure. Barbour's timeless physics [Barbour] takes "possible configurations of the universe and a law" as fundamental and derives time — but the configuration space, the multiplicity of possible instants, is primitive; what is derived is the ordering and the arrow. The Wheeler-DeWitt equation is timeless and time is recovered semiclassically — but on a space of geometries given at the outset. Even recent attempts to derive time from quantum entanglement have been observed to "use time in the set-up." Across both literatures — philosophy of time and the physics of timeless approaches — the same result holds: accounts derive the ordering of moments, or the arrow among them, from structures that already contain the multiplicity of moments. None derives the multiplicity. The domain-existence question is, in the words this note's Section 5.2 reached independently, not one that any known account answers.

Difference and repetition — Deleuze. The pairing of this note's two axioms — differentiation, and its recurrence — is the explicit subject of Deleuze's *Difference and Repetition* [Deleuze], the major modern treatment of the difference/repetition pair. The engagement matters because Deleuze's account of repetition is, at first sight, in tension with this note's Axiom 2. Deleuze insists repetition is "difference without a concept" — *not* the recurrence of the same but the production of novelty; and he ties repetition to the synthesis of time, citing Hume's observation that a repeated series (AB, AB, AB…) itself

produces something new, a habit, an expectation. One might read this as a claim that recurrence is not a primitive but a synthesis, and that time is its product — which would make Axiom 2 a derived structure rather than an axiom. It does not. Deleuze's syntheses operate on an *already-given manifold* of instants: the AB,AB series already has its multiple terms; the passive synthesis of habit contracts a succession that is already there. What Deleuze derives — habit, expectation, the directedness from past through present to future — is temporal *ordering and arrow*, the Section 5.1 ordering level. He does not derive, and does not claim to derive, the bare existence of more than one instant. Deleuzian repetition therefore stands with Wassermann and Barbour: it presupposes the multiplicity and works upon it. It corroborates the no-go rather than threatening it. Two points of contrast are worth stating plainly. First, this note uses "recurs" in Axiom 2 in a deliberately thinner sense than Deleuze's "repetition": Axiom 2 asserts only that differentiation is instantiated more than once over, taking no position on whether the recurrence reproduces the same or produces novelty — that richer question is downstream of the bare domain-existence Axiom 2 asserts. Second, where Deleuze makes difference the sole ontological primitive and repetition a synthesis, this note finds two posits, not one, precisely because — as Section 5.2 establishes — the recurrence cannot be synthesized from the difference without a premise that already contains it.

5.4 The status of succession — a second axiom

Succession is therefore neither evidential nor derivable. But it is also not an independent axiom of the same kind as rung 1. The honest characterization is more specific.

Differentiation can be instantiated more than once over. There is differentiation *within* a configuration — this-here against that-there, the content of a single frame. And there is differentiation *across moments* — this-moment against that-moment. The principle invoked is the same in both: tokened differentiation. What the second instance adds is not a new principle but a new fact — that the principle is instantiated *recurrently*, more than once over, and not only within a single frame.

That fact is a second axiom. It is tempting to call it a mere parameter — a setting on the one axiom — or to absorb it by saying the one axiom is simply "applied across multiple domains." Both moves fail, and fail informatively. A posit that can be neither derived nor eliminated, and must be asserted, is an axiom, whatever it is called; "parameter" understates it. And "applied across multiple domains" is grammatically singular but contentually dual: application presupposes the existence of what one applies across, so the phrase silently carries the very second posit it appears to dissolve. The honest count is two.

The two axioms are:

- **Axiom 1.** *Tokened differentiation occurs.* Concrete differentiated content is instantiated; there is some this-not-that. This is the evidence of Section

2. It is indubitable — its denial is itself a tokened differentiation — and it is conceptually primitive: even the vocabulary needed to state the second axiom presupposes it.

- **Axiom 2.** *Differentiation recurs.* The principle of Axiom 1 is instantiated more than once over — not only within a single configuration but across a further axis. The framework's temporal domain is the minimal such recurrence; the configurational and temporal instances are the two minimum witnesses, and the actual count of axes is settled downstream by derivation, not fixed here. What Axiom 2 asserts is bare recurrence — that the principle holds more than once over — and nothing more: it does not by itself supply an ordering of the recurrent instances, a causal topology, or the specific dynamics φ; these are either left open (ordering, §5.1) or derived downstream from structure the framework independently has (causal topology and arrow, §5.2 inventory). The floor is two posits at the granularity stated, and the dynamical content the downstream physics requires enters through later, separately-stated structural conditions rather than through Axiom 2's semantics.

The two axioms stand in a definite relationship, and stating it precisely is part of the result. Axiom 2 **presupposes** Axiom 1: there can be no recurrence of differentiation without differentiation, and the very notion of an "axis" or "domain" along which the recurrence runs is itself a differentiation-structure, unstatable without Axiom 1's content. But Axiom 2 is **not derivable from** Axiom 1: that a differentiation occurs does not yield that it recurs — this is the no-go of Section 5.5. The relationship is therefore one-directional presupposition without entailment. Axiom 1 is first — conceptually and epistemically. Axiom 2 is genuinely additional. They are two, ordered, and not reducible to one.

A note on what is *not* claimed. The two axioms are not unrelated posits. The principle they invoke — tokened differentiation — is single; Axiom 2 is that single principle asserted to recur. The unity is real, but it is a unity of *theme*, not of *count*: one principle, two axioms. To let the thematic unity be reported as a unity of count — "one axiom" — would be to conceal the second posit behind the first, which is the move this distillation exists to expose, not to perform.

5.5 The result stated as a no-go

The content of Sections 4 and 5 can be stated as a single result. *Given Axiom 1, "tokened differentiation occurs," the rungs relationality, proper-part, dynamics-existence, determinism, finiteness-per-moment, and bijectivity follow as lemmas; the recurrence of differentiation — Axiom 2, equivalently the existence of more than one moment for the dynamics to act across — does not, and cannot, follow.* The negative half is a no-go theorem of modest scope but real content: it identifies, exactly, the one place where the framework's foundation makes a second commitment, and it shows that the second commitment cannot be discharged by any derivation from the first — it must be asserted as an independent axiom. The no-go is specifically about the *recurrence* of differentiation (the existence of

the temporal domain), not about the *ordering* of moments within that domain; the latter is a separate and possibly tractable question (Section 5.3), and the framework takes no position on it here.

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6. THE TWO-AXIOM OBSERVATION BASE

6.1 Statement

The foundation of the framework, distilled, is two axioms:

> **Axiom 1.** *Tokened differentiation occurs.* — Concrete differentiated content is instantiated: there is some this-not-that. Indubitable; conceptually primitive.
>
> **Axiom 2.** *Differentiation recurs.* — The principle of Axiom 1 is instantiated more than once over, across a further axis. The temporal domain is the minimal such recurrence; configurational and temporal instances are the two minimum witnesses, the actual count of axes being settled downstream by derivation.

We call this two-part foundation the **two-axiom observation base**. The two axioms are ordered: Axiom 2 presupposes Axiom 1 (recurrence of differentiation requires differentiation, and the very notion of an axis along which recurrence runs is a differentiation-structure), but Axiom 2 is not derivable from Axiom 1 (Section 5.5, the no-go). The relationship is one-directional presupposition without entailment.

The foundation comprises two axioms, not one axiom plus a domain-multiplicity *parameter*: a posit that can be neither derived nor eliminated, and must be asserted, is an axiom, so the count is two. The two axioms invoke a *single principle* (tokened differentiation), Axiom 2 being that principle asserted to recur. The unity is thematic, not numerical — one principle, two axioms — and is recorded as the relationship between the axioms, not as a reduction of their count.

Everything else in the foundational layer is a lemma: relationality, the locus, the proper-part structure $V \subsetneq S$, the existence of a dynamics, determinism, finiteness-per-moment, and bijectivity. The hidden-sector conditions C1–C4 are selection conditions, not lemmas, and are so treated.

6.2 What the two-axiom form buys

The two-axiom statement is more honest than "a single axiom" and more precise than "several assumptions," and both improvements are substantive.

Against "a single axiom": that phrasing, unqualified, suggests the framework's foundation is a single indubitable fact. It is not. It is one indubitable fact (Axiom 1) *plus* one further commitment (Axiom 2) that is not indubitable — a

world without recurrence, a single tenseless configuration, is coherently conceivable, merely a world with no dynamics and no physics. The two-axiom form states this. It does not weaken the framework: a foundation whose second, non-indubitable commitment is named, isolated, and shown by a no-go theorem to be irreducible to the first is more defensible than a foundation whose commitments are distributed invisibly through a definition. The framework need not claim "one axiom"; the true claim — two axioms, ordered by presupposition, provably not collapsible — is the stronger thing to be able to say, because it comes with a theorem rather than a slogan.

Against "several assumptions": the eight-rung ladder might suggest the framework rests on eight independent posits. It does not. Six of the eight are lemmas. The framework rests on two axioms, and the two-axiom form states *that* — the economy is real, and naming it at the correct count is part of stating it honestly.

6.3 Relation to other minimal claims

The two-axiom observation base is sometimes compared, in discussion, to other candidate "bedrock" claims. The comparisons are clarifying. They concern Axiom 1 — the indubitable, conceptually primitive member — since it is Axiom 1 that occupies the "bedrock" role.

It is not "the universe exists." That asserts a totality; Axiom 1 asserts only tokened differentiation, and the totality (S, the unexhausted whole) is recovered as a lemma (rung 3), not assumed.

It is not "a being exists" — in particular not any claim of the form "X exists" for X a substance with properties. The distillation refuses substance-nouns throughout: it rejected "I exist" and "an observer exists" for importing a persisting subject, and rejected "occurrence grounds itself" for importing a self-grounding thesis. Axiom 1 asserts an occurrence and is silent on what, if anything, underlies it. It is ground-neutral by construction, and that neutrality is deliberate: it is the same discipline that rejected attempt (5) of Section 5.2.

It is not "something exists" simpliciter. That is too thin to be the evidence (Section 2.2): bare undifferentiated existence is not what is given. Axiom 1 asserts *differentiated* content — more than bare existence — which is both what is actually given and, as Section 4 shows, exactly enough structure to generate the lemmas.

Axiom 1 belongs to the family of claims that cannot be doubted because doubting instantiates them — the family of the Cartesian *cogito*. It differs from the others in that family by being maximally disciplined about claiming *only* the indubitable floor and not the structure that the tradition has usually built onto it without notice. Axiom 2, by contrast, makes no claim to indubitability; it is a substantive posit, defended not as undeniable but as the minimal further commitment without which there is no dynamics and no physics, and shown by the no-go to be irreducible to Axiom 1.

6.4 Relation to the landscape: generality, selection, and what they presuppose

A natural question, given that the framework elsewhere derives a gauge structure and a selection principle, is whether the two axioms might themselves be understood on the model of the string-theory landscape: the laws are not unique but range over a vast space of consistent possibilities — gauge or moduli freedom in the loose sense — and *which* possibility obtains is fixed, in our case, by anthropic selection, since only observer-permitting possibilities are observed. Could Axiom 2 — the recurrence of differentiation, the existence of the temporal domain — be such a selected feature: gauge in general, specific in our anthropic case?

It could not, and the reason is the same no-go established in Section 5, arriving from a new direction. Anthropic selection on a landscape operates on features *within an already-constituted physics*: the cosmological constant, the compactification, the couplings each take different values across the landscape, but every point of the landscape is a temporally-extended world with a configuration space and a dynamics. The landscape is a landscape *of* differentiated, temporally-extended things. Anthropic selection picks among such worlds; it cannot pick whether there is a temporal domain at all, because the selection apparatus — a measure over possibilities, the counting of observer-permitting cases — already presupposes the temporal domain. A "world" with no recurrence of differentiation has no dynamics, no vacua, nothing to be a point of a landscape and nothing for a selection measure to weigh. The proposal "Axiom 2 is an anthropically-selected landscape variable" therefore relocates the second commitment into the meta-level structure of the landscape rather than removing it: the landscape itself presupposes Axiom 2. This is the same outcome as the "parameter" and "condition" framings of Section 5.4 — a setting is a setting *within* a space of possibilities, and the space of possibilities cannot be constituted without the very structure the setting was meant to make optional. It may be recorded as a further attempt — alongside the six of Section 5.2 — examined and found to presuppose what it would explain.

String theory itself bears on the two axioms only in this presuppositional way. It is formulated on a worldsheet with a time coordinate, in a target space with a time dimension, over a configuration space that is a multiplicity; it inherits both Axiom 1 and Axiom 2 and questions neither. String theory is not a competing account of the foundation discussed here — it operates well downstream of it, and whatever its landscape is, it is a landscape of worlds in which both axioms already hold.

The landscape comparison is useful as a proposal about observation-relative equivalence and selection, but the complete substratum gauge group is **not** yet classified. OI's exact finite observational representation and its separately scoped observer-admission results remain distinct from the conditional cubic branch, its Bell/gravity completion, and the open all-partitions equivalence.

These distinctions constrain the philosophical analogy rather than make it a proved complete physical selection mechanism.

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7. THE INTUITION IN PLAIN LANGUAGE

The preceding sections are precise at the cost of being technical. The underlying intuition is simple, and stating it plainly is worthwhile — both because the simplicity is the point and because a reader who holds the intuition can check the technical claims against it.

Begin with the one thing that cannot be doubted. Not "the world is real," not "I exist," not that anything in particular lies behind appearances — those are already guesses about what is behind it. Strip them away. What remains, before any guessing, is this: there is something rather than nothing, and not a blank something but a something *with a difference in it*. This, not that. The fact that things are not all the same.

That is the whole starting point. Not "a universe exists" — that already assumes one great container holding everything. Not "a God exists" — that assumes a thing that has properties and does things. Just: *difference shows up*.

It is the right floor, and not a thinner or thicker one, for a specific reason. Thinner — "something exists" — is too thin: plain existence with no difference in it is not anything you are ever actually given, and nothing could be aware of it or doubt it. Thicker — "a universe," "matter," "I" — is smuggling: each sounds obvious only because one already lives inside the assumption, and each, once questioned, dissolves back to the thin fact underneath, that there is difference.

Now the second piece, the piece that took the most care to get right. There are two ways difference can show up, and they are genuinely different. Difference *across a scene*: this thing here is not that thing there — two things side by side, in one frozen snapshot. And difference *across time*: this moment is not that moment — one thing, but later.

One might hope the second follows free from the first — that "there is difference" hands you time automatically. It does not. A single frozen photograph is full of difference — light here, dark there, edges, shapes — and contains no time whatsoever; nothing in it is before or after anything else. So "there is difference" gives the photograph. It does not give the movie. To get the movie — time, change, one-thing-then-another — one must add something: that difference runs in the time direction too, not only across the scene. And that addition cannot be squeezed out of the first claim. Six ways of trying to squeeze it were attempted; every one smuggled time in through a word that already had time hidden inside it.

So the honest foundation is two things of two different kinds. The first is *known*: there is difference — it cannot be doubted, because doubting is itself a telling-

apart. The second is *assumed*: the difference runs in the time direction as well. The framework does assume the second — plainly our world has time — but it is honest that this is a thing put in, not a thing found.

In one sentence: *"there is difference" is what we know; "the difference includes time" is the one thing we assume; physics is what follows from the two — and being able to point at exactly that one assumed piece is what makes the foundation honest.*

That last clause is the reason the exercise matters. Most attempts at a foundation for physics begin, without noticing, from "there is a world, with things in it, and time passing," and never identify how many separate commitments that involves. The two-axiom observation base says: one thing is indubitable, one further thing is assumed, and here is precisely which is which. A foundation in which the single assumed brick can be pointed at is sturdier than one in which the assumptions are smeared invisibly through everything — not because it assumes less in total, but because it is honest about what it assumes.

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8. OPEN QUESTIONS AND STATUS

Part I is developmental, and several questions are deliberately left open.

The succession no-go. The central negative result — that the existence of the temporal domain is not derivable from tokened differentiation — rests on the exhaustion of six derivation attempts (Section 5.2), corroborated by the independent verdict of the philosophy-of-time literature (Section 5.3). Six is not infinity. The result would be overturned by a seventh attempt that succeeds: a derivation of the temporal domain — more than one moment — from a tenseless differentiation premise that does not, at any step, smuggle a process- or temporal-order predicate, and does not begin from a multiplicity (a set of states, a configuration space) that already contains the domain. The author has not found such a derivation and has given a structural reason to expect none exists — the state-to-process gap is not the kind of gap an argument crosses — but the question is left open for scrutiny. A philosopher of physics is the natural reader to either close it or confirm its standing. The distinction of Section 5.1 should be kept in view in any such scrutiny: a reductive account of the *ordering* of moments (Section 5.3 notes a recent candidate) does not bear on the no-go, which is about domain-existence.

The count, and the relationship between the axioms. Part I settles on two axioms — Axiom 1 (tokened differentiation occurs) and Axiom 2 (differentiation recurs) — related by one-directional presupposition: Axiom 2 presupposes Axiom 1 but is not derivable from it. Earlier formulations that sought a single-axiom statement (a "parameter," a "condition," a principle "applied across domains," a reverse derivation of Axiom 1 from a domain-multiplicity axiom, or an anthropically-selected landscape variable — Section 6.4) were each

examined and rejected; each, on inspection, either understated Axiom 2's status or relocated the second commitment without removing it. The two-axiom reading is the current resting point. It is offered, like the rest of Part I, as developmental: whether the relationship between the axioms admits a still more economical regimentation is left open, but no single-axiom formulation found so far survives scrutiny, and the burden is on any proposed one to show it does not conceal the second posit.

What is not open. Two points are not left open. First, the lemma recoveries of Section 4 — relationality, the locus, the proper-part structure, dynamics-existence, determinism, finiteness-per-moment, bijectivity — are presented as established, given the axiom; they turn on conceptual analysis of "occurs" and "tokened" and are checkable directly. Second, the distillation has no bearing on whether the framework's downstream derivations succeed; it concerns only the formulation of their starting point.

PART II — CONSTRUCTIVE STRUCTURAL REALISM

Part II examines an observation-relative structural-realist proposal. Theorem 23 of [Substratum] supplies conditional outputs of a selected cubic branch; Theorem 24 supplies exact invariances for specified observational carriers and a restricted non-exhaustiveness result. Their conjunction does not establish a unique inverse reconstruction or a complete physical gauge quotient. The argument below separates these established conditional results from the physical selection and equivalence questions that remain open.

9. STRUCTURAL REALISM AS A PROGRAMMATIC POSITION

Ontic structural realism (OSR) holds that what is fundamental, and what physical theory gives us knowledge of, is structure — the network of relations — rather than individuals bearing intrinsic natures. The distinction between OSR and a merely epistemic structural realism was drawn by Ladyman [Ladyman1998], and the ontic position has been developed principally by Ladyman and Ross [LadymanRoss] and by French [French2014], with the relation to quantum physics worked out by French and Ladyman [FrenchLadyman2003] and a recent "effective" refinement given by Ladyman and Lorenzetti [Ladyman-Lorenzetti]. The position emerged from two pressures: the pessimistic meta-induction, which observes that the theoretical objects of successive physical theories are repeatedly discarded while their structural-mathematical content is retained; and the underdetermination of object-level descriptions by structure, sharpest in physics where distinct object assignments (particles-as-individuals

versus particles-as-modes, for instance) leave all relations and all predictions intact.

OSR responds: take the structure as the ontology, and the recurring instability of the object level becomes unsurprising, because the object level was never the bearer of the theory's content. A standing concern for the position, recognized in its own literature, is that "structure" not collapse into "pure mathematics" — French and Ladyman were explicit that OSR is not committed to the world being purely mathematical — and the question of how the structural and the physical are related remains under discussion. The framework's treatment, developed below, bears on exactly this concern: it makes the structure a specified mathematical object while keeping the physical content fixed by empirical inputs, and it is explicit (Section 12) about what is thereby claimed and what is not.

As ordinarily developed — in the work of Ladyman, French, and others — OSR is a *programmatic* position. It is a stance one adopts about physical theory in general, defended by general arguments (the meta-induction, the underdetermination), and it makes a general promise: that for a mature physical theory, the structure could in principle be specified and the object-level residue shown to be surplus. The promise is rarely discharged for any specific theory. What "the structure" is, precisely; which respects of a theory's description are structural and which are surplus; whether the structure is uniquely fixed by the empirical content or itself underdetermined — these are, in the programmatic version, left as the position's horizon rather than its deliverable.

The constructive-structural-realist claim is currently **partial**. The framework exhibits a selected local cubic branch and proves restricted observational invariances, but it does not derive uniqueness of the local substratum from E1–E7/A1–A6, nor a complete gauge quotient. Theorem 24 distinguishes exact fixed-interface coherent-channel preservation, broader population-only invariance, and open all-partitions equivalence; Pair C refutes exhaustiveness of the restricted coherent-channel generators. Bell-composite structure remains conditional on H-Bell.

10. CONDITIONAL LOCAL RECONSTRUCTION AND ITS OPEN UNIQUENESS QUESTION

The forward construction maps a selected finite deterministic model and observer interface to observable laws. The converse from observations to a **unique** local substratum has not been proved: Theorem 23 records conditional lattice/gauge outputs on the chosen nearest-neighbor cubic/H-link branch, and the stronger inverse theorem needs a valid physical branch selector and observational equivalence classification.

Scope of the current reconstruction. Theorem 23 gives conditional outputs of the $d = 3$ nearest-neighbor cubic branch under H-link ($m = 1$), the

octahedral representation, amplitude-scale gauge and separate [SM] matter assumptions. It does not uniquely select this branch from E1–E7/A1–A6 or recover a full substratum gauge class. The reference nearest-neighbor graph is Bell-local over sufficiently separated wings under the stated interventions; a common Bell-violating, no-signaling and curvature-stable completion is an open H-Bell obligation. The massless action-scale relation also requires H-slope and the horizon/frame conditions [GR §3.2].

Two features of this theorem matter methodologically, independently of whether each forced output survives empirical scrutiny.

The first distinction is the direction of the result. Theorem 23 derives stated local outputs after fixing the nearest-neighbor cubic branch and its physical hypotheses. The converse from empirical inputs to a unique local structural realization remains unproved, even within A1–A6 without further selection assumptions. H-Bell adds a separate completion problem; limiting a uniqueness claim to the local sector does not supply the missing converse.

The second distinction concerns the inputs. A1–A6 restrict the candidate class, but they are not sufficient by themselves to select the branch used in Theorem 23. Its outputs additionally consume the chosen wave-rule carrier, H-link with $m = 1$, the octahedral representation, amplitude-scale gauge and the relevant matter hypotheses. The gravitational relation uses separate horizon conditions. These premises determine the scope of each deduction; naming A1–A6 does not turn it into a uniqueness theorem within that class.

The empirical inputs E1–E7 are idealized structural facts; finite data do not establish them outright. Granting them exactly still does not make Theorem 23 an inverse uniqueness result: the theorem retains the selected branch as an additional premise. Finite-data model discrimination, local-branch selection and compatibility with Bell composites are distinct questions. The quantitative comparisons in [SM §7] bear on their classified derivation chains and do not replace a proof of microscopic uniqueness.

11. OBSERVATIONAL EQUIVALENCE: INVARIANCES, OBSTRUCTIONS AND OPEN CLASSIFICATION

A structural-realist interpretation must distinguish branch-relative mathematical outputs from a **proved observational equivalence class**. Theorem 23 currently supplies conditional selected-branch results, whereas the general classification of dynamics preserving physically accessible data is open. Proposition 24.2 supplies a separate Lean-proved fixed-carrier block-word rigidity theorem when all word traces and the spanning premise are given; the visible channel family alone does not supply them.

Current gauge results. For a fixed visible/hidden interface, a uniform hidden permutation preserves full coherent channels, while a hidden permutation

depending on the visible fibre preserves rooted population probabilities but can alter coherences. An exactly decoupled normalized ancilla also preserves coherent channels wherever the partial trace is defined. Pair C shows these restricted coherent-channel moves are **not exhaustive**. Across all admissible observer partitions, the equivalence and completeness classification remain open. Alphabet-size results and graph equivalences transfer only their explicitly stated structural content [SM §2.7; Substratum §4], not unconditional equality of all observable channels.

The decisive distinction is **which observable carrier is being compared**. The earlier use of Stinespring uniqueness to infer a single conjugation of the entire generated unitary from equality of all visible channels fails (Pair C). Hidden-unitary block-word rigidity is proved only with the stronger equality-of-word-traces and spanning premises (Substratum Proposition 24.2). That result does not by itself show that these premises follow from OI-accessible data or that every equivalence is generated by four transformations. No complete classification follows for the all-partitions relation.

Finite Stinespring uniqueness and the current fixed-carrier block-word theorem remain valuable constructive ingredients, but neither classifies the full substratum observational quotient. The analogy with the QFT problem of inequivalent representations should be read as a proposed research comparison rather than a resolved finite counterpart. A physical choice of observable carrier and a valid inverse theorem are still required before claims about the exact observational surplus can be made.

The structural-realist programme has conditional calculations on a selected branch and exact invariances for specified carriers. It remains incomplete as an inverse reconstruction and as a classification of the full physical quotient. Pair C shows that the restricted fixed-interface coherent-channel generators are non-exhaustive; all-partitions equivalence requires its own formulation and proof.

12. WHAT THE CONJUNCTION ESTABLISHES, AND WHAT IT DOES NOT

The methodological yield of Parts II is the conjunction, so it is worth stating exactly.

The two theorems presently provide a selected-branch structural result and explicitly restricted invariances, not a complete construction of a unique observational equivalence class. This is a substantive methodological limitation: a physical specification of what the observer reads, a proof selecting the local branch, and the classification of all surviving redundancies must be supplied independently. Bell-composite uniqueness remains open as a further question.

Three limits constrain the structural-realist interpretation. The local outputs depend on the additional branch and physical premises stated in [Substratum Theorem 23]. The observational transformations of Theorem 24 are carrier-

specific and do not give a complete quotient: Pair C refutes exhaustiveness of the restricted coherent-channel moves, while all-partitions classification remains open. Finally, a common Bell-compatible physical completion has its own H-Bell obligation. The construction supplies conditional results and exact invariances, not a completed instance of inverse ontology reconstruction.

The Doplicher–Roberts reconstruction theorem provides a mathematical-physics precedent for recovering field and gauge structure from appropriately specified observable data [Halvorson]. It does not prove the OI inverse problem. In OI, a selected local model supplies conditional lattice/gauge outputs, but recovering a complete substratum equivalence class from observations requires an additional selector and an observational classification. The analogy identifies a form of research question, not an established correspondence between the two reconstruction results.

The methodological moral, which Part IV will generalize, is already visible. Once the structure is pinned and the surplus is classified, the question "but which substratum is the real one?" stops being a deep unanswered question and becomes a question shown to have no content. That dissolution — not the answering of the question but the demonstration that it was ill-formed — is the pattern the framework applies again, in Part III, to the no-go theorems of quantum foundations.

PART III — PER-THEOREM EVASION OF NO-GO RESULTS

Part III develops the framework's third methodological commitment: that its relation to the no-go theorems of quantum foundations is governed by a single structural feature rather than by four separate escapes. The Bell and Wigner's-friend analyses summarized here are established in the framework's Main paper; the Kochen–Specker and reality-theorem cases are developed in this part, each as a chain of reasoning running on results the technical papers already establish. The unifying account — that the framework's two-level, trace-out architecture is what each theorem's assumption runs aground on — is the present paper's synthesis, and the developmental status of each case is flagged accordingly.

13. THE NO-GO THEOREMS AS A FAMILY

A deterministic-substratum account of quantum mechanics must answer to the no-go theorems of quantum foundations. These results — Bell's theorem [Bell], the Kochen–Specker theorem [KochenSpecker], the reality theorems in the Pusey–Barrett–Rudolph family [PBR], the Frauchiger–Renner theorem [FrauchigerRenner] — each show that some intuitively natural class of accounts of quantum mechanics cannot reproduce its predictions. A framework that posits a deterministic substratum underneath quantum mechanics is, on its face,

exactly the kind of account these theorems constrain, and a reader encountering such a framework is right to ask, immediately, how it survives them.

The framework's answer could in principle be piecemeal: a separate manoeuvre for each theorem, each manoeuvre exploiting whatever assumption that particular theorem happens to leave open. Piecemeal answers are methodologically weak — they invite the suspicion of ad hoc construction, a different escape hatch cut for each prison — even when each individual answer is technically correct. The methodological claim of this part is that the framework's answer is not piecemeal. The four theorems are met by one structural feature of the construction, and the appearance of four separate evasions is dissolved into a single observation about levels of description.

The single observation is this. Each of these no-go theorems assumes some property — locality and factorizability, or noncontextuality, or a form of state-realism, or the universality of a single quantum description — and shows that no account having that property at the level of its hidden or underlying variables can reproduce quantum mechanics. The framework has two levels of description — a hidden deterministic substratum and an emergent quantum description — joined by the trace-out, the marginalization of the substratum dynamics over the unobserved hidden sector that produces the reduced observable law — universally admitting the quantum representation — in the first place. The framework's relation to the whole family of theorems is mediated by that one piece of machinery, already at its core, and fixed before any no-go theorem is considered.

The trace-out separates the two levels, but the four no-go theorems do **not** share one automatic escape pattern, and the manuscript should not pretend that they do. Bell is sharper: for a response-complete deterministic completion, a finite-range Bell graph together with a setting-independent pre-setting ensemble gives a Bell-local model and $|S_{\mathrm{CHSH}}| \leq 2$. The framework therefore cannot use the trace-out, screening, or indivisibility as an escape from Bell; it retains measurement independence and takes ontic parameter dependence for Bell-violating substrata, with operational no-signaling kept separately. Kochen–Specker constrains the noncommuting operational observable algebra, so its inheritance is conditional on the coherent operational lift. PBR depends on a preparation-independence postulate for the underlying variables; OI does not posit that postulate, but graph causal cones alone do not prove its failure or a particular quantum tensor factorization. Frauchiger–Renner turns on the relation between observer-relative operational descriptions and the substratum. What the four cases genuinely share is the need to keep substratum statements, fixed-basis representation statements, and full operational-QM statements at their correct levels. The trace-out is the common architecture, not a universal theorem-disabling device.

14. THE FOUR CASES

A status note, stated before the cases and not buried within them. Bell's case now has two deliberately separated layers. Spatial OI gives an exact graph causal cone for finite-range substrata. But when the completion is response-complete and deterministic, combining that pointwise cone with one measurement-independent pre-setting ensemble gives a Bell-local hidden-variable model and the classical ceiling $|S_{\mathrm{CHSH}}| \leq 2$. The framework therefore does not use screening plus graph locality as its route to quantum Bell correlations: it declines the measurement-dependent/superdeterministic branch and takes ontic parameter dependence for Bell-violating substrata, while retaining operational no-signaling. The Barandes–Hasan–Kagan Tsirelson result remains a statement about the weaker stochastic causal-local indivisible description, not about the deterministic completion. Kochen–Specker is conditional on the common coherent noncommuting instrument lift. The PBR discussion is likewise conditional at the composite level: OI does not postulate unconditional preparation independence, and co-embedding permits shared hidden data, but the exact composite operational structure is not supplied by the graph cone alone. Frauchiger–Renner is an interpretive consequence of partition-relative reduced descriptions. These different statuses are kept separate below.

Bell's theorem. Bell's theorem binds on the framework because the substratum configuration is a hidden variable. The dependency graph proves a finite-propagation statement for any substratum whose Bell-relevant couplings are finite-range: until the remote setting's cone reaches the local readout region, the remote setting cannot change the local response through that graph. If, in addition, a pre-setting variable fixes both wings' responses to every setting and its ensemble is independent of the settings, determinism turns the model into an ordinary local hidden-variable decomposition, hence $|S_{\mathrm{CHSH}}| \leq 2$. That is the Bell ceiling theorem of Main §3.3. Quantum Bell violation therefore cannot be attributed simultaneously to that pointwise cone and measurement independence.

The framework's chosen branch is ontic parameter dependence: the Bell-relevant substratum has coupling range exceeding the wings' separation, so one wing's response can depend on the remote setting at the microstate level even though the averaged outcome distribution is operationally no-signaling. The alternative is ontic measurement dependence; Main records but does not adopt it, with the necessary bound $S \leq \min\{4, 2 + 4\sqrt{2 \ln 2\, I_{\mathrm{ont}}}\}$. The stochastic causal-local indivisible construction can obey Tsirelson because its causal-independence condition is weaker than Bell factorization; that result is retained at the stochastic layer only. Unrestricted globally controlled OI realizations can produce PR-box behavior, so no-signaling, hidden memory, and indivisibility are all insufficient by themselves to select the quantum correlation set. *(Established: graph causal-cone theorem for finite-range substrata, Bell ceiling for the screened deterministic completion, and the measurement-dependence bound; adopted Bell branch: ontic parameter dependence with operational no-signaling.)*

The Frauchiger–Renner theorem. Frauchiger–Renner shows that quantum theory cannot consistently describe its own use by an agent who is themselves a quantum system, if one assumes a single universal quantum description applying to all agents at once. The framework's Main paper addresses this through its treatment of the Wigner's-friend structure. In the framework, observer identity is partition-gauge: Friend and Wigner correspond to different choices of the visible/hidden partition of the same substratum, and each choice yields its own internally consistent reduced description. Friend's claim to a definite outcome and Wigner's description of Friend as a traced-out subsystem are statements in different reduced descriptions, not competing claims about the substratum. At the substratum level all parties have definite configurations; the apparent paradox is a covariance property — predictions are covariant under choice of observer partition.

In the levels language: Frauchiger–Renner assumes a single universal quantum description. The framework denies exactly this at the relevant level — there is no single quantum description, there are multiple partition-relative reduced descriptions, each obtained by a different trace-out of the one substratum. The universal description the theorem needs as its target does not exist in the framework; what exists universally is the substratum, which is not a quantum description at all. The theorem's assumption fails not by contradiction but because the framework's structure does not contain the object the assumption is about. *(Established: framework Main paper.)*

The Kochen–Specker theorem. Kochen–Specker shows that no noncontextual hidden-variable assignment — one assigning every observable a definite value independent of which compatible set of observables is measured alongside it — can reproduce quantum mechanics. The framework's relation to it is a short *conditional* argument on results the framework already has. The framework reproduces the quantum statistics at the fixed-basis transition layer: the characterization theorem delivers a full equivalence there ($S \iff D \iff Q_{\mathrm{fb}}$, Main §3.4). Kochen–Specker is a theorem *about* the operational theory — it states that the observable algebra of quantum theory in dimension three or higher admits no noncontextual value assignment — and that algebra of noncommuting observables belongs to the operational bridge Main states as open (§3.2, §3.4). The inheritance is therefore conditional and automatic: if the coherent operational lift reconstructs the standard observable algebra, any account carrying it carries a structure with no noncontextual valuation, and the framework's emergent valuation is necessarily contextual, on pain of contradicting that operational structure. Contextuality is thus not a property the framework must be argued into case by case — it is forced by the equivalence, once the bridge is crossed.

What the framework adds is the *account* of that contextuality, and here the Section 13 architecture does the work. An emergent observable's value is produced by the trace-out, and the trace-out is taken with respect to a partition; measuring a compatible set of observables is a specific apparatus coupling, and

the framework models a measurement context as exactly such a coupling-and-partition configuration (the double-slit treatment of Main §3.4 is the worked instance: a detector couples the system to additional visible-sector degrees of freedom and thereby changes the transition matrix). Which emergent observables are jointly defined, and the value each takes, are features of the reduced description fixed by that partition — not partition-independent functions of the substratum configuration. Contextuality, in the framework, *is* the partition-dependence of emergent valuation: the same partition-relativity Part II identified as gauge. The framework need not reconstruct the Kochen–Specker colouring obstruction itself; that obstruction is inherited intact with the operational theory, once the bridge selects it. The framework is a deterministic hidden-variable theory whose hidden level is a definite substratum configuration but whose emergent valuation is contextual — precisely the kind of hidden-variable theory Kochen–Specker leaves open. *(Established: this paper, on the Main paper's equivalence theorem.)*

The reality theorems (Pusey–Barrett–Rudolph family). The PBR theorem shows that, given a natural assumption about how independently prepared systems combine — the Preparation Independence Postulate, that the joint ontic-state distribution of two separately prepared systems is the product of the single-system distributions — the quantum state cannot be a mere probability distribution over deeper states of reality, but must be ontic. A deterministic-substratum framework is naturally read as making the quantum state epistemic: the state is the observer's representation of incomplete knowledge of the substratum configuration. The framework's relation to PBR turns on the Preparation Independence premise and on the still-separate composite operational lift; it is therefore conditional rather than already closed by the fixed-basis equivalence.

The Preparation Independence Postulate requires an unconditional product distribution for the ontic states of independently prepared systems. OI does not impose that postulate at the substratum level: two preparation devices embedded in one substratum may share preparation-relevant hidden or boundary data even when no later dynamical signal passes between their laboratories. This is enough to show that PBR cannot simply be imported as a theorem about bare OI without first establishing its preparation-independence premise. The point should not be strengthened into a theorem that graph locality itself yields a particular conditional product distribution or a quantum tensor-product state: the graph gives causal cones, while the exact composite/tensor operational structure belongs to the coherent operational lift still stated separately in Main. Consequently the framework's relationship to PBR remains conditional at the composite layer. If a proposed OI completion were shown to satisfy PBR preparation independence together with the other hypotheses, it would inherit PBR's conclusion; if it does not, the theorem is outside that completion's scope.

15. WHY THE UNIFICATION IS NOT AD HOC

The natural objection to any framework that "evades" several no-go theorems is that the evasions are opportunistic — that the framework has simply looked, for each theorem, at which assumption can be broken cheapest, and broken it. Part III's claim is that this objection does not apply, and the reason is structural rather than rhetorical.

The four cases are not four versions of one theorem-disabling trick. They share one bookkeeping architecture — substratum, reduced description, and trace-out — but each no-go theorem constrains a different object and must be checked separately. Bell is the clearest warning against over-unification: Bell factorizability fails for the observed quantum correlations themselves, and a response-complete deterministic completion cannot recover those correlations from graph locality plus measurement independence. The adopted substratum is therefore ontically parameter-dependent, while the emergent statistics remain no-signaling. Frauchiger–Renner concerns the universality of one quantum description across observer partitions; PBR concerns preparation independence at the ontic composite level; Kochen–Specker concerns noncontextual valuations in the noncommuting operational algebra. The common factor is the discipline of locating each premise at the correct level, not a claim that the trace-out automatically makes every forbidden premise true above and false below.

The test of non-ad-hocness is whether the response to the theorems was available before the theorems were examined, or constructed afterward to fit. Here it was available before: the two-level trace-out architecture is the framework's account of how the quantum-representable reduced description arises at all — it is Part I's and Part II's subject, the construction itself — and it was not built to answer Bell or Frauchiger–Renner. That the same architecture, unmodified, answers the no-go family is a consequence drawn from it, not an adjustment made to it. This is the methodological content of the per-theorem evasion claim: not that the framework escapes four theorems, but that it does not have four escapes — it has one structure, and the four "escapes" are one fact about that structure, viewed four times.

The honest status, restated: the Bell and Frauchiger–Renner cases are developed in the framework's Main paper and summarized here faithfully; the Kochen–Specker and PBR cases are developed in Section 14 of this paper as *conditional* chains that run on results the framework's technical papers already establish — the fixed-basis representation equivalence for the Kochen–Specker case, and that equivalence together with the spatial-Markov and tensor-product structure of Main §3.3–3.4 for the PBR case — with the inheritance crossing to the operational theory once the selection bridge reconstructs the noncommuting observable algebra (Main §3.2). Two cases are thus developed and two conditional, and a reader is invited to check the two chains and their shared condition directly. The methodological claim of Part III stands on all four: the framework's relation to the no-go family is governed by one architectural feature, and each

of the four cases exhibits it.

PART IV — PHYSICS MODULO GAUGE

Part IV draws the three preceding commitments into a single methodological proposal: that the well-posed question for fundamental physics is not "what is fundamental?" but "what is the equivalence class of structures that produce the observations, and what are its representatives?" This part argues the proposal as a general methodology, not as a feature peculiar to the framework's particular construction.

16. THE QUESTION THAT DOES NOT CLOSE

The realism debate in the philosophy of physics is organized around a question: what is fundamental? Realists answer with an inventory — particles, fields, spacetime points, a wavefunction on configuration space — and divide over which inventory is correct. Anti-realists decline the inventory and divide over what to say instead. The debate is durable, and its durability is itself a datum. A question that has resisted resolution for as long as this one has — through every major change in physical theory, with the candidate inventories turning over each time while the debate's structure persists — invites the suspicion that the question is not hard but ill-formed.

The preceding parts suggest a methodological replacement for unqualified substratum realism: ask which observable carrier is actually fixed and which microscopic differences are indistinguishable *under it*. The strongest reconstructed local branch is currently conditional; its candidate gauge transformations do not furnish an exhaustive classification for fixed-interface coherent channels, and the all-partitions case remains open. This is not evidence that every unobservable degree of freedom is already known to be gauge.

The methodological distinctions concern premises, observable structures and representations. For transformations proved to preserve a specified carrier, measurements of that carrier cannot choose among the related descriptions. It does not follow that the empirical evidence fixes one complete physical equivalence class or that every candidate representative is redundant. Theorem 23 leaves the local inverse open, and Theorem 24 does not supply a complete physical classification. The philosophical question must therefore be assessed relative to the equivalence actually established.

17. THE WELL-POSED REPLACEMENT

If "what is fundamental?" is ill-posed, it should be replaced, not merely abandoned. The replacement the framework's methodology proposes is: *what is the equivalence class of structures that produce the observations, and what are its representatives?*

The methodological questions can be stated without assuming their answers. Which local models are compatible with the specified observations, and which transformations preserve the chosen observable carrier? Theorem 23 constrains a selected branch but does not answer the first question uniquely. Theorem 24 exhibits exact coherent-channel and population invariances but does not classify every representative; its restricted coherent-channel generators are not exhaustive. H-Bell adds an independent common-completion question. A complete physical equivalence class therefore remains a target of reconstruction.

Call the methodological aim *physics modulo a specified observational equivalence*. A complete quotient is a possible endpoint of rigorous reconstruction, not a premise OI has already delivered. At present, finite-horizon laws admit universal representations, the selected lattice branch has conditional structural consequences, and different observable carriers lead to different notions of microscopic redundancy. The philosophical value lies in explicitly separating those facts rather than assuming a four-family gauge quotient.

The methodological proposal can be compared with analyses of gauge surplus and observational equivalence by Healey, Wallace and others [Healey]. The relevant mathematical comparison must name the observable structures and the transformations acting on them. OI has partial fixed-interface invariances, a refuted restricted-generator completeness claim (Pair C), and an open all-partitions classification; their distinction is part of the foundational research programme, not an already closed representation theory.

This methodology supplies an avenue for reinterpreting the realism debate and Wigner's puzzle, but it does not presently prove a bidirectional reconstruction of the physical substratum from observed data. Theorem 23 constrains a selected local branch; H-Bell and the all-partitions gauge classification remain separate open problems. Any ontological claims made from these results must retain those limitations.

A mathematical-universe interpretation could use observer admission as a selection question: which specified models and interfaces support the memory/readback conditions of [Main]? This does not make Theorem 23 a selector of one physical substratum or define a measure over all mathematical structures. The local branch and physical inputs remain additional premises, and the all-partitions observational classification is open. The philosophical proposal must retain these limits.

18. THE PROPOSAL AS A GENERAL METHODOLOGY

The claim of this part is not merely that the Incompleteness framework happens to admit a physics-modulo-gauge reading. It is that physics modulo gauge is a methodology — a way of posing and answering foundational questions — that other programs in fundamental physics could adopt, and against which they can be measured.

The proposal can be located against a long-standing open problem. Hilbert's sixth problem [Hilbert] called for the axiomatization of physics — the treatment of physics, in those parts ready for it, with the rigour of an axiomatic mathematical theory. More than a century on, the problem is generally regarded as only partially addressed, and a recognized obstacle is that the primitive notions of physical theories — "body," "force," and their successors — tend to be, in the phrase of the problem's modern commentators, pregnant with theory rather than genuinely primitive. The methodology proposed here bears on that obstacle directly. Its discipline is to identify the genuine primitives by stripping the theory-laden ones away — Part I's distillation of "observation occurs" to a two-axiom base is exactly this move, performed at the foundational layer — and then to ask, of the structure those primitives plus the empirical inputs determine, the two well-posed questions of physics modulo gauge. The proposal is, in this sense, a partial and specific response to Hilbert's sixth problem: not an axiomatization of physics entire, but a method for stating what a particular foundational program commits to, with the primitives exposed and the gauge freedom classified.

Stated generally, the methodology has three steps. First, for a candidate fundamental theory, identify the empirical inputs and ask whether they determine a structure uniquely — whether a reconstruction result is available. Second, where uniqueness holds only up to some equivalence, classify the equivalence explicitly: exhibit the gauge group, and prove the classification exhaustive, so that the surplus is bounded and known. Third, having the class and its gauge group, decline the question of which representative is real, and certify that declension with the gauge classification itself — the representatives are inter-transformable by observable-preserving maps, so the question has no empirical content.

A program that carries out these three steps has, on this view, said everything there is to say about what its theory commits to. A program that cannot carry out step one — that cannot exhibit its structure as determined by the evidence — has not yet earned a realist reading of that structure, whatever its other merits. A program that carries out step one but not step two — that has a structure but no classification of its gauge freedom — has an unbounded surplus and cannot yet say which of its commitments are genuine. The methodology is, in this sense, a standard: it specifies what a foundational program must exhibit before its ontological claims are well-posed, and it does so in a way that is checkable, because each step is the kind of thing that is done by a theorem or not done at all.

Physics modulo a specified observational equivalence is a methodological programme. OI contributes a forward construction, conditional selected-branch outputs and exact invariances for named carriers. The inverse branch selector, complete observational classification and full inter-level maps are unfinished. A different programme could formulate analogous reconstruction and equivalence questions for its own observables; OI's partial results do not certify that those steps have been completed in general.

The framework partially instantiates the proposed methodology: it supplies finite-law representations, conditional branch reconstruction and specified interface invariances, while the physical selector and complete observational classification remain open. These results provide concrete tests of the reconstruction programme rather than a completed example of all three steps. Whether the methodology generalizes to other foundational programmes is a further research question.

19. GAUGE FREEDOM AND OBSERVATIONAL INCOMPLETENESS

This section is the most recent addition to the paper and the least settled; it is developmental even by the standards of the rest, and is offered as a proposal for scrutiny. It connects the methodology of Parts II–IV to the framework's core physical thesis — the incompleteness of observation — and argues the two are, on a precisely delimited class, one principle.

A proper instantaneous readout can omit state information without making that information permanently inaccessible. In a known reversible model, retained readings can identify the full initial and current state when the supported record fibre is a singleton [Substratum Proposition 24.5]; the two-reading swap is an exact example. State uncertainty, finite resource limits and redundancy under a proved observational invariance are therefore distinct. Only the last identifies descriptions relative to the specified preserved data.

Observational redundancy is relative to the information a specified interface supplies. A uniform hidden-factor relabelling preserves the coherent channel family under Theorem 24's hypotheses; fibre-dependent hidden relabellings preserve rooted populations without necessarily preserving coherences. For either carrier, a measurement depending only on the preserved data cannot distinguish the related descriptions. This observation does not imply that all hidden information or all microscopic differences are gauge.

This identification must be delimited, because not every observational incompleteness is gauge, and the cases that are not are as instructive as the cases that are. Consider two ways the framework's embedded observer fails to have a fact.

In the first case, the chosen data are invariant under a demonstrated transformation. Hidden-only relabelling and exactly decoupled ancillary enlargement give examples for coherent channels; the population carrier admits a broader hidden-fibre relabelling. A model with coupled additional hidden dynamics is not covered by the independent-ancilla result. Calling the demonstrated redundancy gauge does not establish that the complete physical theory contains no further distinctions or that the all-partitions equivalence has been classified.

A quantity can also be definite while remaining unresolved under specified preparation, access, record or computational constraints. Such a limitation requires a resource or indistinguishability argument for the actual model and interface;

proper embedding alone gives no whole-system size lower bound for predicting a visible future. Known reversible models can have predictable readouts, and retained records can identify their state. A resource-limited unresolved fact can be called **computational incompleteness** when that limit has been established; unlike a proved redundancy of descriptions, it does not remove the quantity's content.

The distinction between these two species is, as far as we are aware, not drawn in the existing literature on observational limitation, and drawing it is part of this section's contribution. With it in place, the identification can be stated exactly. *Gauge freedom is identical to gauge incompleteness — the species of observational incompleteness in which the inaccessible question has no answer-bearing content.* It is not identical to computational incompleteness. "Physics modulo gauge," the methodology, and "observational incompleteness," the physics, name one principle restricted to the gauge-incompleteness species: the methodology-side name and the physics-side name of the same fact.

The distinction between formulational and interpretational surplus remains relevant [NguyenTehWells; BradleyWeatherall]. For OI, interpretational redundancy can be asserted only relative to a specified observable carrier and transformations proved to preserve it. Theorem 24 provides examples and a restricted non-exhaustiveness result, not a classification of every transformation preserving all physical observables. A complete ontology quotient cannot be assumed in using the philosophical distinction.

There is a consequence of this identification that is worth stating in closing, because it is the point at which the paper's methodology turns back on the paper's own first part. If gauge freedom is the gauge-incompleteness species of observational incompleteness, then physics modulo gauge — the methodology — is a methodology *for working in the presence of observational incompleteness.* And observational incompleteness, in this framework, is not a special circumstance; it is the condition of any embedded observer, which is to say of any observer at all. The methodology is then not a special technique for a special class of theories. It is the general form that honest physics takes once it is admitted that the physicist is inside the system. Part I found that the framework's own foundation, examined closely, has a feature — whether the underlying contrast is tenseless or temporal — that is gauge: a question the framework can pose and that no observation it could ever make will answer. That the framework's *own foundational description* contains gauge incompleteness is not an embarrassment to be hidden. It is the methodology applying to itself, exactly as it should, if the methodology is correct: the physicist is inside the system even when the system in question is the physicist's own foundation.

The comparison with Deutsch's account of open-ended knowledge [Deutsch] is interpretive. A fixed observation interface can fail to distinguish descriptions related by a proved invariance. A computational limitation, by contrast, can concern a quantity that differs in that carrier but exceeds available resources. The distinction is meaningful without a complete gauge classification; it does

not establish that all hidden or currently inaccessible quantities lack physical content.

Knowledge derived solely from specified observational data cannot distinguish models assigned identical data by a demonstrated comparison. This is a conditional information-access statement. It does not identify a unique full substratum class, and it does not make OI's partial invariances a proof of a general epistemological thesis. Relating them to Deutsch's argument requires keeping the chosen access conditions explicit.

PART V — THE OPERATIONAL METHODOLOGY

Part V develops the methodology operationally without assuming that the substratum gauge group has been completely classified. It asks which quantities can be determined from a specified representative and observer/readout map, and which remain dependent on additional choices. Its two no-go results and the accompanying audits have their own hypotheses and should not be read as proving the missing all-partitions equivalence. Part V remains developmental in the sense described in the paper.

21. THE OPERATIONAL QUESTION

Parts II–IV motivate a methodology of physics modulo *specified* observational equivalences. They do not establish a complete quotient of substratum dynamics: coherent-channel, population-only, and all-partitions comparisons differ, and Theorem 23 currently concerns a conditional selected branch. Part V's operational question thus asks which results can be derived from a given substratum and observation map, and which additionally require a physical selector; it does not assume the missing quotient has already been classified.

The question is non-trivial in both directions. Some quantities the framework would *like* to derive from substrate computation alone turn out to depend on choice of representative; the operational methodology forces these to be classified differently than the framework's first-pass framing would suggest. Other quantities turn out to be substrate-derivable in ways that the framework's first-pass framing did not make explicit; the operational methodology surfaces these as content that had been undercounted. Part V develops both directions through two no-go theorems (§22, §23), the architectural pattern that unifies them with Part III (§24), the systematic audits the technical papers carry out (§25), and the methodological refinements the audits canonize (§26).

The status of Part V as a whole is the same as that of Parts II–IV: the methodology is a proposal, the worked instances are developed results in the framework's

technical papers, and the unification stated here as synthesis is the present paper's contribution. Each section flags its own status more specifically.

22. THE GAUGE-INVARIANCE NO-GO

*Status: conditional operational argument. The invariance logic below is valid for transformations separately established to preserve the selected observational interface; a complete substratum gauge classification is not supplied by Theorem 24. This section does not promote the original four-family completeness route.

The first operational no-go concerns **specified transformations known to preserve the chosen observable carrier**. Because Theorem 24 does not furnish a complete general gauge group, this result must not quantify over an assumed four-generator observational quotient as though that quotient were proved. A substrate-only computation can be assessed for invariance under the particular transformations actually established; any further claim about all physically equivalent substrata requires a separately justified equivalence relation and a proof that the computation descends to it.

The conditional proof is elementary. **Fix** an explicitly specified observable carrier and an explicitly established transformation g preserving every observable of that carrier. Let Q be a function of the underlying dynamics for which $Q(g\varphi) \neq Q(\varphi)$. Any experimental procedure restricted to that carrier gives the same distribution of accessible outcomes before and after g, so it cannot infer both different values of Q uniquely from the common outcome law. A computation performed on one microscopic representative may still return a number, but without independently selected representative information that number cannot be certified as an observer-invariant prediction. The conclusion holds for the stated transformations; it does **not** prove that every difference between substrata is gauge or that the full quotient is known.

The methodological examples below distinguish putative representative-dependent assignments from quantities computed within a selected branch. Their classification as physically gauge-dependent requires testing the *actual* observation map and permitted transformations, not assuming a complete $\mathcal{G}_{\mathrm{sub}}$ in advance. In particular, choosing which experimentally identified generation receives a mixing-angle ratio or mass-order label can introduce empirical information beyond a bare algebraic calculation. The precise role of this choice is testable separately for each prediction.

The operational lesson is to identify the observation map and tested invariances before claiming a microscopic quantity is either gauge-derivable or necessarily empirical. For example, a magnitude relation may follow from a selected representation while the assignment of that magnitude to named physical states uses additional calibration. This distinction is independent of, and does not establish, the still-open classification of all substrata consistent with the same physical observations.

There is **no converse of the form** 'every gauge-invariant function is computable from the substratum' without independent computational, regularity and access assumptions. Invariance is a necessary condition for an observable prediction to be well-defined on the quotient, not a theorem of physical accessibility or algorithmic computability. Reported Cabibbo, Koide, horizon-entropy, MOND and mixing-angle calculations retain their separate branch, calibration, dynamical and empirical-input conditions; they cannot be certified as outputs of a complete substratum gauge quotient by the no-go argument alone.

23. THE CATEGORY-ERROR FINDING

Status: developed result; the type-mismatch is established by direct inspection of the framework's substratum-level structure (see the framework's substratum paper, §2.5, and the technical-paper record more broadly). The methodological reading is the present paper's synthesis.

The second operational result is more pointed than the first. It states that some questions the framework's first-pass framing makes available are not merely gauge-forbidden but ill-typed: the question asks for an object the framework's substratum-level structure does not contain.

The instance through which the finding was identified is the substrate-level Lagrangian. The framework's quantitative content (mass ratios, mixing angles, coupling thresholds) takes Lagrangian form at the *emergent* level: post-trace-out, after the substratum dynamics have been marginalized over the unobserved hidden sector, the visible-sector dynamics admit a description as a quantum field theory with an action functional. The substrate-level dynamics, by contrast, are *not* given by an action functional. The substrate is the deterministic linear bijection $\varphi : S \to S$ of the foundational definition; its dynamics is the iteration of φ, not the extremization of an action. A "substrate-level Lagrangian" is therefore not a thing the framework's substratum-level structure offers; the construction has no place to attach.

The mismatch is type-level rather than calculational. It is not that a substrate-level Lagrangian is hard to compute or that the relevant calculations are beyond current tools; it is that the request itself does not pick out a well-typed object in the framework's foundational ontology. Asking for "the substrate Lagrangian" is structurally similar to asking for "the temperature of a single classical particle" — temperature is a thermodynamic quantity defined on ensembles, and applying it to a single particle is not a difficult question but a category error. The framework's substrate is the bijection on a finite set; Lagrangian field theory is the post-trace-out description that emerges from marginalizing the bijection over the hidden sector. The two live at different levels of description, joined by the trace-out, and the joining is one-directional: emergent Lagrangians come *from* the substrate by trace-out, but there is no inverse map that produces a substrate-level Lagrangian *from* the emergent one.

The methodological point generalizes beyond the Lagrangian case. Any frame-

work with a level distinction — substrate and emergent, fundamental and effective, microscopic and macroscopic — will have constructions natural at one level that do not transfer to the other, and not all such non-transfers are blocked by gauge invariance in the §22 sense. Some are blocked by type-mismatch: the construction is not defined at the level it is being asked of. The methodological discipline is to recognize the difference. A gauge-forbidden quantity exists at the relevant level but cannot be picked out by substrate-only computation; a type-mismatched construction does not exist at that level at all. Conflating these blocks two different kinds of inquiry under the same label and obscures what the framework's substrate-level commitments actually are.

The category-error finding has a corrective consequence for the framework's own research program. The framework's technical record contains a sequence of attempts at substrate-level closure of certain quantitative predictions — attempts that constructed candidate substrate-level Lagrangians and looked for the framework's coefficients to fall out of those Lagrangians. The category-error finding establishes that such attempts were not failing for calculational reasons; the targets they aimed at were not well-typed objects in the framework's foundational ontology. Re-classifying the relevant quantities as either equivalence-class invariants (substrate-derivable in the sense of §22) or bijection-specific (selected by empirical input) — rather than as targets for substrate-Lagrangian closure — is the corrected research direction. The reclassification is internal to the framework, but the methodological pattern it instances is general: when a foundational program finds itself repeatedly stuck on a construction, the first check should be whether the construction is well-typed at the level it is being asked of.

24. THE ARCHITECTURAL PATTERN: PART III EXTENDED

Status: synthesis; the architectural pattern is articulated in Part III as a unification of the four no-go cases of quantum foundations. The extension to the §22 and §23 results is the present part's synthesis claim, offered as a proposal in the same sense Part III's unifying formulation is offered.

Part III argued that the framework's relation to the four no-go theorems of quantum foundations is not four separate evasions but one architectural feature applied to four distinct theorems. The feature is the substrate/emergent split joined by the trace-out: two levels of description, the substrate level deterministic and global, the emergent level the marginalization of the substrate over the unobserved hidden sector, and the trace-out the operation that produces the second from the first. Bell, Kochen–Specker, the Pusey–Barrett–Rudolph reality theorems, and Frauchiger–Renner each have an assumption that the framework declines, and in each case the declension is located by the substrate/emergent architecture rather than by a custom-built evasion for the theorem in question.

The operational conclusion of §22 concerns a procedure restricted to specified

preserved outcome data: it cannot infer two different values of a quantity from the same data law. This does not constrain a computation supplied with a microscopic representative or a richer intervention interface, and invariance alone does not prove derivability. The §23 question is separate: a proposed Lagrangian requires a defined carrier, dynamics and limiting map before a substrate-level calculation can address it.

Stated as a thesis, the pattern is that the framework's claims and limitations can be organized by the substrate/emergent architecture and the trace-out joining the levels. Part I specifies the foundational structure; Part II gives conditional selected-branch outputs and demonstrated invariances for specified carriers, while the full substratum equivalence classification remains open. Each proposed no-go application must therefore identify its own premises and accessible data. The claimed unifying methodology is to locate those questions within the specified architecture, rather than to infer their answers from an already completed gauge classification.

The unification has a verifiable consequence. If the architectural pattern is right, then any further no-go encounter the framework runs into should also be locatable by the substrate/emergent split. The pattern is therefore falsifiable in a specific sense: a no-go result the framework cannot meet by locating where the assumption lands relative to the architecture would be evidence that the unification is incomplete and that a custom move is required. Part III and Part V have, to date, located every no-go encounter the framework has taken up by this single move; whether that record continues to hold under further scrutiny is part of what is offered for testing. The unification is a working claim, not a settled one.

25. THE OPERATIONAL AUDITS

Status: the audits themselves are developed in the framework's technical record; their methodological reading as systematic application of §22 to the framework's own content is the present part's synthesis.

The §22 gauge-invariance no-go is methodologically empty if not applied. A framework that proves a no-go theorem about substrate-only computation and then continues to claim quantities are substrate-derivable when they are in fact bijection-specific would have stated a result without using it. The framework's technical record applies the §22 result to its own content through systematic audits, which is part of what the operational methodology of this part means by "operational."

For each prediction, an audit must identify the actual observable carrier, input data, proved invariances and derivation. A quantity that varies between models with the same permitted data is not identifiable from those data alone. A quantity invariant under the demonstrated transformations may still lack a derivation or be uncomputable within stated resources. Classification as derived,

conditional, retrodictive or empirical input follows the explicit argument and its premises, not invariance by itself.

Two audits in particular illustrate the discipline. A physics-sector audit covering the technical papers on the Standard Model and gravitational sectors classified twenty-nine predictions, finding approximately fifteen substrate-derivable (i.e., equivalence-class invariant), four substrate-derivable in magnitude with bijection-specific directional content, and ten emergent-level mechanism-distinctive items that the framework's first-pass framing had not separated from the substrate-derivable cluster. A cross-domain audit covering chemistry, biology, and computational complexity classified thirty-four cross-domain claims under the same criteria; a mid-audit correction caught a generous-grading pattern and revised the substrate-derivable count downward, separating mechanism-distinctive items from substrate-derivable ones in line with §22's distinction. The numerical findings of both audits are in the framework's technical record; what is methodologically important here is that the audits ran on the same standard the framework's operational methodology proposes for foundational programs in general.

A third audit applied the same standard recursively to a previous expansion of the substrate-derivable count. The expansion had identified eight further candidates not covered by the prior audits and reclassified them as substrate-derivable; the recursive audit found that three of the eight survive a strict-criteria filter as approximately partially independent substrate-derivable items (with correlation discounts reflecting that they share machinery with already-counted items), four are mechanism-distinctive in §22's sense (distinctive in the framework's *explanation* for outcomes that the framework shares with the standard account), and one is framing rather than predictive. The recursive correction is methodologically the cleanest of the three findings: a framework that catches its own corrective moves inflating, and walks the inflation back to what survives strict criteria, is exhibiting the discipline the operational methodology recommends to others.

The resulting taxonomy separates derivation under stated hypotheses, identifiability from specified outcome data, empirical selection and explanatory interpretation. The operational result of §22 supplies a necessary invariance condition for data-restricted identification, not a converse deriving every invariant or a universal prohibition on microscopic computation. The complete gauge classification remains open, so each application names the transformations actually proved to preserve its chosen carrier.

The audits' general lesson concerns what the methodology asks of its users. A framework that proposes physics modulo gauge as its methodology takes on an obligation to apply the methodology to its own claims, and to revise those claims where the operational classification requires it. The framework's technical record shows that obligation being met more often than not — including in the recursive case above, where the obligation extended to revising the framework's own audits — though not always in the first pass. The methodological

refinements that this discipline canonizes are the subject of §26.

26. THE OPERATIONAL REFINEMENTS

Status: the refinements stated here are derived from the framework's audit experience and are offered at the generality the operational methodology requires. The discipline they specify is canonical within the framework's technical record at multiple-instance strength; whether it generalizes to other foundational programs is the proposal-status claim.

The audit experience canonizes three methodological refinements that the operational methodology recommends as standards for any foundational program of the kind Parts II–IV describe. Each is stated below at the generality the methodology requires, with the framework-specific instance noted for reference but not relied on for the generality of the claim.

First refinement: verify before recording. When a finding adjusts a framework's count of substrate-derivable content — whether upward (more items than the prior count suggested) or downward (fewer items than the prior count suggested) — the finding should be subjected to strict-criteria filtering before being recorded as the new count. Positive findings inflate at first pass: the natural pattern is to count items that look as if they should qualify, and to discover on closer inspection that several of them do not meet the strict criteria. The discipline of running the strict-criteria filter on every count-change finding cuts the inflation, often by a non-trivial fraction. The framework's audit record contains eight instances of this pattern across one session of work, at strengths sufficient to canonize the refinement at the framework's internal level. The general claim is that any foundational program whose methodology proposes a categorized inventory will face the same inflation pressure, and the same strict-criteria filtering is the corrective.

Second refinement: recursive scrutiny. A corrective finding — one that catches an inflation in a prior finding — is itself subject to the same inflation pressure. The framework's audit record contains one instance of this case (a corrective finding that itself inflated the framework's count of substrate-derivable content by a factor of three at first pass, and that walked back to a strict-criteria count after a second-order audit). The refinement is provisional in the framework's own canonization scheme — second-order corrections are rare enough that one instance does not establish the pattern at the same strength as the first refinement — but the methodological point is general: self-correction does not stop at one pass, and the strict-criteria discipline applies recursively. A framework that catches the inflation in its own corrections is exhibiting the methodology more sharply than a framework that catches only the inflation in its original findings.

Third refinement: the observable/derivable distinction. A foundational program's open-question list is not a uniform category; it contains questions that fall into at least four distinct buckets. Some questions concern *observable* con-

tent the framework does not yet predict — open empirical questions in the standard sense. Some concern content that is *derivable* from the framework's existing structure but has not yet been derived, requiring further calculation rather than further structure. Some concern content that is *structurally permitted* (consistent with the framework's commitments) but whose specification is gauge — equivalent representatives produce different specifications, none of which has empirical content. Some concern content that is *structurally forbidden* — gauge-incompatible specifications, ruled out at the level of the framework's structure. The four buckets ask different things of further work, and mixing them in a single "open questions" list obscures what each kind of question requires for closure. The methodological refinement is to distinguish the four buckets explicitly. The framework's open-question list, when classified by this distinction, is shorter than the unclassified list suggests, because some apparent open questions resolve to "structurally permitted but gauge" — not closures, but acknowledgments that the question has no answer-bearing content.

The three refinements share a structure. Each takes a category that a foundational program's working vocabulary collapses — the category of "predictions," the category of "self-correction," the category of "open questions" — and disaggregates it into sub-categories that ask different things and admit different kinds of further work. The methodology's discipline, throughout Parts II through V, has been the disaggregation move: track what is committed, distinguish it from what is not, and at each level find that the distinction the puzzle was failing to track is what makes the puzzle tractable. The refinements of §26 are that move applied to the methodology itself: track what kind of operational claim is being made, distinguish derivation from inflation from declension, and the methodology gains the same precision it asks the framework's content to display.

27. CONCLUSION

This paper has articulated the methodology of the Incompleteness framework through five commitments, and it is worth stating, in closing, what holds them together.

Part I separates the framework's indubitable observation premise from the additional temporal recurrence commitment. Part II presently gives conditional selected-branch structural outputs, some exact invariances, and a refuted restricted-generator completeness claim, not a finished full-gauge structural realism theorem. Bell-composite completion, the physical branch selector, and all-partitions observational equivalence remain open. The methodology is therefore a rigorous programme of separation and testing, rather than an already complete inverse reconstruction.

The methodological requirement is to distinguish stated premises, conditional deductions, observed quantities and demonstrated representational redundancies. At the structural level, Theorem 23's selected-branch outputs must be separated from a physical inverse selector, and Theorem 24's carrier-specific in-

variances from a complete classification. At the operational level, an invariance argument quantifies only over transformations proved to preserve the chosen data. These distinctions permit precise questions without claiming that the remaining scientific and interpretive questions have already been closed.

That discipline is replicable, which is the paper's reason for stating it separately from the framework's physics. The framework's quantitative claims — the Standard Model retrodictions, the entropy coefficient, the neutrino prediction — stand or fall on their own evidence, in the technical papers. The methodology stands or falls on whether the discipline just described is sound and general. The two assessments are independent, and a reader may find the methodology worth adopting whatever they conclude about the physics, or worth rejecting whatever they conclude about the physics. We have marked, throughout, where the paper's claims are developmental: the foundational distillation of Part I is recent and offered for scrutiny; the structural theorems of Part II are summarized from the technical papers at the fidelity of faithful synthesis; the four no-go cases in Part III are all developed results, two of them developed in this paper rather than in the technical papers and flagged for the referee accordingly; the generality of the Part IV methodology is offered as a proposal, not a result; the identification of gauge freedom with the gauge-incompleteness species of observational incompleteness, in Section 19, is offered explicitly as a proposal for scrutiny; and Part V is more recent still — its two no-go results are developed in the framework's technical record but their methodological reading as part of one architectural unification with Part III is offered as a synthesis, the systematic audits are themselves developed but the discipline they instantiate is proposed for use beyond the framework, and the three operational refinements are canonical within the framework's record at multiple-instance strength while their generality to other foundational programs is the proposal-status claim. What is not developmental is the discipline itself — track what is committed, distinguish it from what is not — and the claim that, applied to the foundations of physics, that discipline is what the framework's methodology consists in.

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This paper articulates, for a philosophy-of-physics readership, the methodological posture developed across the Incompleteness framework's technical papers. Part I develops the framework's foundational commitment; Parts II and III draw on structural results from the framework's substratum and main papers.

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